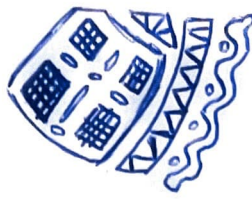


Markov decision problem



e.g.

The travelling salesman
The traffic light at intersection
Selling Magazines

Dynamic Programming

$V_t(x)$: optimal expected profit that can be made with t days to go and current inventory level of x

On Friday (using red convention for inventory level x)

$$V_1(x) = \max_{y \geq x} (p E[\min(y, D_1)] - C_1(y-x)) = \max_{\alpha \in \mathcal{A}_x} r_1(x, \alpha)$$

$$x^+ = \max(x, 0)$$

On Thursday

$$V_2(x) = \max_{y \geq x} (p E[\min(y, D_2)] - C_2(y-x) + E[V_1((y-D_2)^+)])$$

$\vdots$

$$V_t(x) = \max_{y \geq x} (p E[\min(y, D_t)] - C_t(y-x) + E[V_{t-1}((y-D_t)^+)])$$

with $V_0 = 0$

$$= \max_{y \geq x} (p (\sum_{k=0}^y k P(D_t=k) + y P(D_t \geq y)) - C_t(y-x) + \sum_{k=0}^y P(D_t=k) V_{t-1}(y-k) + P(D_t > y) V_{t-1}(0))$$

Optimal profit is $V_5(0)$

$\vec{\mu} = (\mu_1, \dots, \mu_5) = (3, 4, 4, 4, 6)$ Demand

$\vec{c} = (0.9, 1.0, 1.1, 1.2, 1.5)$ Cost

$p = 3.5$ Selling Price.

$V_5(0) = 43.6926 \text{ €}$ Optimal profit.

Discrete Markov chains

A Markov chain is a stochastic process indexed on the integers that has the Markov property

If $\{X_t\}_{t \in \mathbb{N}}$ is a Markov chain then

$$P(X_{t+1} = y \mid X_t = x, X_{t-1} = x_1, X_{t-2} = x_2, \dots) = P(X_{t+1} = y \mid X_t = x) = p(x, y)$$

Let $\mathcal{X}$ denote the state space of X_t , then:

$$\sum_{y \in \mathcal{X}} p(x, y) = 1$$

NOTE $P(X_t = x) \stackrel{\text{def}}{=} \pi_t(x) = \sum_{y \in \mathcal{X}} p(y, x) \pi_t(y)$

Simplifying assumptions:

01/04/26

① $|\mathcal{X}| < \infty$ (finite state space)

Def ^{PATH} A sequence of states $z_0, \dots, z_k \in \mathcal{X}$ s.t.
 $p(z_0, z_1), \dots, p(z_{k-1}, z_k) > 0$ is called a path
 from z_0 to z_k of length k

② There is at least one state x such that there
 is a path from any state $y \in \mathcal{X}$ to x . (the RECURRENT STATE)
 (UNICHAIN Markov chain)

③ The GREATER COMMON DENOMINATOR of all path lengths from x
 to itself is 1.
 (APERIODIC Markov chain)

Since $|\mathcal{X}| = n$ we can number states. Let $p(x, y) = p_{xy}$

Transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & & \vdots \\ \vdots & & \ddots & \\ p_{n1} & \dots & & p_{nn} \end{bmatrix}$$

$$\pi_t = (\pi_t(1), \dots, \pi_t(n))$$

Now $\pi_{t+1} = \pi_t P$ and $\pi_t = \pi_0 P^t$

NOTE

TRANSITION MATRIX COMMUTE

Thm
 Under assumptions (1-3) independent of initial distribution π_0

from $\pi_t = \pi_*$ exists and satisfies $\pi_* = \pi_* P$, $\pi_* e^T = 1$

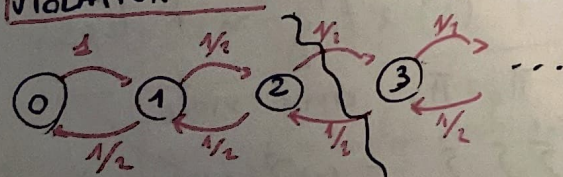
π_* is stationary if $\pi_0 = \pi_*$ then $\pi_t = \pi_* \forall t$

$$\uparrow$$

$$e = (1, \dots, 1)$$

EX. VIOLATIONS

VIOLATION OF 1:



Let $\mathcal{X} = \mathbb{N}_0$ suppose $\pi_0(0) = 1$, $p(0,1) = 1$ and $p(x, x+1) = p(x, x-1) = \frac{1}{2}$

$$\pi_2(2) = \frac{1}{2} = \pi_2(3) = \frac{1}{2} \Rightarrow \pi_2(x) = \pi_2(x+1) = c \forall x$$

suppose $c > 0 \rightarrow$ NOT A DISTRIBUTION

$$c > 0 \rightarrow \sum_{x \in \mathcal{X}} \pi_2(x) = c + c + \dots \neq 1$$

VIOLATION OF 2

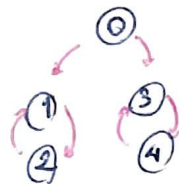
Let $\mathcal{X} = \{1, 2\}$



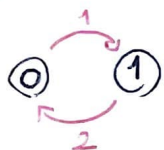
$$P = \Pi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\pi_x = \pi_0$ so NOT INDEPENDENT OF π_0 .

← ALSO NOT UNICHAIN.



VIOLATION OF 3



$$\pi_0 = (1, 0)$$

$$\pi_1 = (0, 1)$$

$$\pi_2 = (1, 0)$$

→ Because if I start in 0, I can visit 0 every even number of periods $\Rightarrow$ periodic.

$\pi_x = \lim_{t \rightarrow \infty} \pi_t$ does not exist unless $\pi_0 = (\frac{1}{2}, \frac{1}{2})$

When π_x exists it is also the long run occupancy distribution

$$\pi_x(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{I}_{\{X_t = x\}} \quad \text{a.s.}$$

Lecture 2

Discrete Markov Reward Chain

Let $r: \mathcal{X} \rightarrow \mathbb{R}$ be the "reward" (cost) of visiting state x .

Then $\mathbb{E}[r(X_t)]$ is the expected reward in period t and

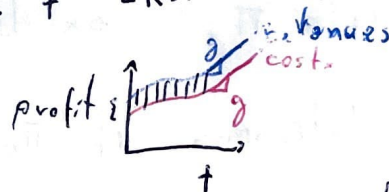
$\mathcal{X}$: state space

$$g = \lim_{t \rightarrow \infty} \mathbb{E}[r(X_t)] = \sum_{x \in \mathcal{X}} \pi(x) r(x) = \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E}\left[\sum_{k=1}^t r(X_k)\right]$$

Develop "Poisson" equation

$$\text{Let } V_T(x) = \mathbb{E}\left[\sum_{t=0}^{T-1} r(X_t) \mid X_0 = x\right] =$$

$$= \sum_{t=0}^{T-1} \sum_{y \in \mathcal{X}} P^t(x, y) r(y)$$



$P^t(x, y)$: probability of jumping from x to y in t steps

Suppose we start at the stationary distribution $\pi_0 = \pi_x$ and note:

$$\sum_{x \in \mathcal{X}} \pi_x(x) V_T(x) = \sum_{x \in \mathcal{X}} \pi_x(x) \sum_{t=0}^{T-1} \sum_{y \in \mathcal{X}} P^t(x, y) r(y) = \sum_{t=0}^{T-1} \sum_{y \in \mathcal{X}} \sum_{x \in \mathcal{X}} \pi_x(x) P^t(x, y) r(y) = \Delta$$

EXPECTED RETURN AFTER T PERIODS

PROB OF START IN x RETURN IN y

because we are assuming stationary distribution

$$\pi_*(y) = \sum_{x \in \mathcal{X}} \pi_*(x) P^+(x, y)$$

$$\Delta = \sum_{t=0}^{T-1} \sum_{y \in \mathcal{X}} \pi_*(y) r(y) = g^* T$$

Let $V(x) = \lim_{T \rightarrow \infty} (V_T(x) - gT)$ Relative value function
Bias

(cost) growth starting in steady state
 growth starting in state x

Difference in total reward from starting in x vs starting in steady state.

$$V_{T+1}(x) = V_T(x) + \sum_{y \in \mathcal{X}} \pi_T(y) r(y) \quad \text{with} \quad \pi_0(x) = 1$$

As $\pi_T \xrightarrow{T \rightarrow \infty} \pi_*$ and $\sum_{x \in \mathcal{X}} \pi_*(x) r(x) = g$

$$V_{T+1}(x) = V_T(x) + \underbrace{\sum_{y \in \mathcal{X}} \pi_*(y) r(y)}_g + \sum_{y \in \mathcal{X}} (\pi_T(y) - \pi_*(y)) r(y) \quad (*)$$

"back ward recursion" (separate period $T+1$)

Note also by recursion

$$V_{T+1}(x) = r(x) + \sum_{y \in \mathcal{X}} p(x, y) V_T(y) \quad (**)$$

"Forward recursion" (separate period 0)

Combining the two:

$$V_T(x) + g + \sum_{y \in \mathcal{X}} (\pi_T(y) - \pi_*(y)) r(y) = r(x) + \sum_{y \in \mathcal{X}} p(x, y) V_T(y)$$

Subtracting gT from both sides and taking $T \rightarrow \infty$.

$$\lim_{T \rightarrow \infty} (V_T(x) - gT) + g + \sum_{y \in \mathcal{X}} \lim_{T \rightarrow \infty} (\pi_T(y) - \pi_*(y)) r(y) =$$

$$= r(x) + \sum_{y \in \mathcal{X}} p(x, y) \lim_{T \rightarrow \infty} (V_T(y) - gT)$$

$V(y)$ by def

Solve recursion (*) for large x
 Solve Poisson Equations if the transition matrix is reasonable

$$\Rightarrow V(x) + g = r(x) + \sum_{y \in \mathcal{X}} p(x, y) V(y) \quad \forall x \in \mathcal{X}$$

POISSON EQUATION(S)

Equations (one for each state $x \in \mathcal{X}$) in $V(x)$

g is known, and the equations are linear in V
 Solution is unique up to a constant. (bc. $\sum_y p(x, y) = 1$)

So we add one more equation to enforce the interpretation "relative to v^* "

$$\sum_{x \in X} \pi_x^*(x) V(x) = 0$$

Markov decision Process (MDP)

Now there is a Policy $R \rightarrow X \rightarrow A$

Assumptions: $|X| < \infty$

$\forall$ policy R there is at least one recurrent state
($\forall$ path induced by $\forall$ policy there must be a r.s.)

"Uncertain MDP"

$\forall$ policy R , the induced Markov chain is Aperiodic.

$p(x, a, y) \rightarrow$ "taking decision a "
 $v(x, a) \rightarrow$, $a = R(x)$ decision given by policy

Total reward accrued after t period if start in x and follow policy R .

$$V_t^R(x) = E_R \left[\sum_{s=0}^{t-1} v(X_{t+s}, R(X_{t+s})) \mid X_0 = x \right]$$

We seek

$$\arg \min_R \lim_{T \rightarrow \infty} \frac{V_T^R(x)}{T}$$

"Average cost-rate criterion"

Eg. X is inventory, A is the ordering decision, $v(x, a)$ is the revenue-cost given the starting state x (i.e. cost for unsold inventory, missed sales etc...)

For one day the boss is away and you get to implement the best policy that optimizes reward for the day and leaves it in a good state. He will then keep following his policy R .

$$\begin{aligned} \min_{a \in A_x} v(x, a) + \sum_{y \in X} p(x, a, y) V^R(y) &= \\ = \min_{a \in A_x} v(x, a) + E_a [V^R(X_{t+1}) \mid X_t = x] \end{aligned}$$

"Equally weighted by using Average cost rate criterion". Alternatives discounted.

Algorithm Policy Iteration

- Take some policy R (your boss policy)
- Compute g^R and $V^R(x) \forall x \in X$ by solving Poisson Eq.
- Determine and improve policy R'

$$R'(x) = \arg \min_{a \in A_x} \left\{ v(x, a) + \sum_{y \in X} p(x, a, y) V^R(y) \right\}$$

If $R' = R$: STOP $\rightarrow$ Policy is optimal. If $R' \neq R$, set $R = R'$ and go to ②.

(EXPECTED RETURN AFTER T PERIODS)

$O(N^3)$ complexity

Improvement theorem

Let g and $V(x)$, $x \in \mathcal{X}$ be given numbers

Suppose that the policy R' satisfies the "Poisson Inequality"

$$v(x, R'(x)) - g + \sum_{y \in \mathcal{X}} p(x, R'(x), y) V(y) \leq V(x) \quad (*)$$

For each $x \in \mathcal{X}$. Then the average cost rate of policy R' satisfies

proof Multiply both sides of $(*)$ by $\pi_{*}^{R'}(x)$ and sum over x

$$\underbrace{\sum_{x \in \mathcal{X}} \pi_{*}^{R'}(x) v(x, R'(x)) - g}_{(A)} + \underbrace{\sum_{x \in \mathcal{X}} \pi_{*}^{R'}(x) \sum_{y \in \mathcal{X}} p(x, R'(x), y) V(y)}_{(B)} \leq \sum_{x \in \mathcal{X}} \pi_{*}^{R'}(x) V(x)$$

Under stationarity $\pi_{*}^{R'}$, $(A) = g^{R'}$, $(B) = \sum_{y \in \mathcal{X}} \pi_{*}^{R'}(y) V(y)$

$$\Rightarrow g^{R'} - g + \sum_{y \in \mathcal{X}} \pi_{*}^{R'}(y) V(y) \leq \sum_{x \in \mathcal{X}} \pi_{*}^{R'}(x) V(x) \Rightarrow g^{R'} \leq g \quad \square$$

CONVERGENCE

• Because $\mathcal{X}$ and $\mathcal{A}$ are finite, the number of policies I can go through is finite, and because g can only improve, the algorithm will converge. (Uniqueness depends on the form of v).

Then Optimality equations

Let $V^*(x)$, $x \in \mathcal{X}$ and g^* be given numbers such that

$$V^*(x) + g^* = \min_{a \in \mathcal{A}_x} \left\{ v(x, a) + \sum_{y \in \mathcal{X}} p(x, a, y) V^*(y) \right\} \quad \forall x \in \mathcal{X}$$

Then $R^*(x) = \arg \min_{a \in \mathcal{A}_x} (v(x, a) + \sum_{y \in \mathcal{X}} p(x, a, y) V^*(y)) \quad \forall x \in \mathcal{X}$ is an optimal policy, and g^* is the optimal cost-rate.

proof By def $V^*(x) + g^* = v(x, R^*(x)) + \sum_{y \in \mathcal{X}} p(x, R^*(x), y) V^*(y) \quad \forall x \in \mathcal{X}$

And $V^*(x) + g^* \leq v(x, a) + \sum_{y \in \mathcal{X}} p(x, a, y) V^*(y) \quad \forall x, \forall a \in \mathcal{A}_x$

Let $\bar{R}$ be any policy. By Improvement th: $g^{\bar{R}} \geq g^*$

Since R^* obtains a lower bound on the cost of any policy, it is an optimal policy and g^* is the opt. cost rate. $\square$

$\mathcal{A}_x$ set of actions that can be taken from state x

Algorithm Linear Programming formulation to solve MDP

Encode a policy π as follows:

$$f_{x,a} = P(A_t = a | X_t = x) \quad \forall t \rightarrow \text{allows randomization, } P \text{ stationary in time}$$

indep of t action taken at time t , given that I am in x .

$$\pi_x = \lim_{t \rightarrow \infty} P(X_t = x) \quad \text{under the policy encoded by } f$$

Depend on t

$$z_{x,a} = \lim_{t \rightarrow \infty} P(X_t = x, A_t = a) = f_{x,a} \pi_x \quad (*)$$

NOTE THAT $\pi_x = \sum_{a \in A_x} z_{x,a} \quad (**)$

Note also that the cost-rate of the policy encoded by f is given by:

$$g = \sum_{x \in X} \sum_{a \in A_x} z_{x,a} v(x,a)$$

The balance equations of the MC induced by policy " f " are:

$$\pi_x = \sum_{y \in X} \sum_{a \in A_y} \pi_y f_{y,a} p(y,a,x) \quad \forall x \in X$$

Using $(*, **)$:

$$\sum_{a \in A_x} z_{x,a} = \sum_{y \in X} \sum_{a \in A_y} z_{y,a} p(y,a,x)$$

Any set of $z_{x,a}$ that satisfy the balance equations and normality $(\sum_{x \in X} \sum_{a \in A_x} z_{x,a} = 1)$ correspond to a policy whose the steady

state probabilities are given by $(**)$ and the policy

$$f_{x,a} = \frac{z_{x,a}}{\sum_{b \in A_x} z_{x,b}}$$

Therefore if we allow randomized policies, we can find an optimal policy by solving the following LP

$$\min g = \sum_{x \in X} \sum_{a \in A_x} z_{x,a} v(x,a)$$

$$\text{s.t. } \sum_{a \in A_x} z_{x,a} = \sum_{y \in X} \sum_{a \in A_y} z_{y,a} p(y,a,x) \quad \forall x \in X$$

$$\sum_{x \in X} \sum_{a \in A_x} z_{x,a} = 1, \quad z_{x,a} \geq 0 \quad \forall x \in X \quad \forall a \in A_x$$

Note there are $|\mathcal{X}|$ basic variables, one for each state.
Also at optimality $\Rightarrow$ optimal policy will not be randomized.

▷ NOTE it also works for "weakly unichain", i.e. only the **OPTIMAL POLICY IS UNICHAIN**

By the improvement theorem and the optimality equations (OE) 31/03/2026 there is a solution to the following inequalities in $V(x), x \in \mathcal{X}, g$:

$$V(x) \geq r(x, a) - g + \sum_{y \in \mathcal{X}} p(x, a, y) V(y)$$

We seek to maximize this lower bound (LB). This gives the following linear program (LP)

$$\begin{aligned} \max_{g, V(x)} \quad & g \\ \text{s.t.} \quad & V(x) \geq r(x, a) - g + \sum_{y \in \mathcal{X}} p(x, a, y) V(y) \quad \forall x \in \mathcal{X} \\ & \forall a \in A_x \\ & g, V(x) \in \mathbb{R} \quad \forall x \in \mathcal{X} \end{aligned}$$

The optimal solution to this LP gives a solution to the (OE),
i.e. $g_{LP}^* = g_{OE}^*$

This LP is the DUAL LP of the previous one.

Algorithm **Value iteration** (use python)

- suppose you wish to control your MDP over a finite horizon
- suppose there is, at the end of the horizon, a **TERMINAL REWARD** for ending in state x , $V_0(x), x \in \mathcal{X}$
- Let $V_t(x)$ be the optimal total reward with t periods to go until the end of the horizon starting in state x .

$$V_{t+1}(x) = \min_{a \in A_x} \left\{ r(x, a) + \sum_{y \in \mathcal{X}} p(x, a, y) V_t(y) \right\} \quad \forall x \in \mathcal{X}$$

- Intuition, for t sufficiently large (like 500 years), V_t should look like the average optimal value function V^* up to some constant.

$R_t(x) = \argmin_{a \in A_x} \left\{ r(x, a) + \sum_y p(x, a, y) V_t(y) \right\}$
should be a near optimal policy.

Value Iteration Algorithm $\epsilon > 0$ (e.g. 10^{-3})

→ step 0: set $V_0 = 0$, $n = 1$, compute:

→ step 1: For each state x compute:

$$R_n = \operatorname{argmin}_a \left\{ r(x, a) + \sum_{y \in \mathcal{X}} p(x, a, y) V_{n-1}(y) \right\}$$

you only need to store R_n

→ step 2: compute bounds on optimal long-run cost-rate.

$$m_n = \min_{x \in \mathcal{X}} \left\{ V_n(x) - V_{n-1}(x) \right\} \leq g^*$$

$$M_n = \max_{x \in \mathcal{X}} \left\{ V_n(x) - V_{n-1}(x) \right\} \geq g^{R_n}$$

→ step 3: check stop criterion

If $0 \leq M_n - m_n \leq \epsilon m_n$ stop

R_n is near optimal $\frac{g^{R_n} - g^*}{g^*} < \epsilon$

otherwise set $n \leftarrow n+1$ and go to step 1.

Th) STOP CRITERION

$$\forall n \in \mathbb{N}, \quad m_n \leq g^* \leq g^{R_n} \leq M_n$$

PROOF
PART 1 Since $m_n = \min_{x \in \mathcal{X}} (V_n(x) - V_{n-1}(x))$ we also have

$V_n(x) \geq V_{n-1}(x) + m_n \quad \forall x \in \mathcal{X}$. Combine this with:

$$V_n(x) \leq r(x, R(x)) + \sum_{y \in \mathcal{X}} p(x, R(x), y) V_{n-1}(y) \quad \forall x \in \mathcal{X}$$

To find that for any policy R (by Impr. Th.)

$$V_{n+1}(x) + m_n \leq r(x, R(x)) + \sum_{y \in \mathcal{X}} p(x, R(x), y) V_{n-1}(y) \quad \forall x \in \mathcal{X}$$

Application of the Impr. Th. now gives, for any policy R that

$m_n \leq g^R$. It applies also to the optimal policy g^* so

$$m_n \leq g^*$$

PART 2 By def.

PART 3 Like part 1

Th) Improved BOUNDS

The lower bound m_n is nondecreasing in n and the upper-bound M_n is nonincreasing:

$$m_{n+1} \geq m_n, \quad M_{n+1} \leq M_n$$

By def of R_n

$$(*) V_n(x) = r(x, R_n(x)) + \sum_{y \in \mathcal{X}} p(x, R_n(x), y) V_{n-1}(y) \quad \forall x \in \mathcal{X} \quad \forall n \in \mathbb{N}$$

For an arbitrary R we have:

$$(**) V_n(x) \leq r(x, R(x)) + \sum_{y \in \mathcal{X}} p(x, R(x), y) V_{n-1}(y) \quad \forall x \in \mathcal{X} \quad \forall n \in \mathbb{N}$$

Take $n=K$ in $(**)$ and $n=K+1$ in $(*)$ and $R=R_K=R_{K+1}$. Subtract $(*)$ from $(**)$

$$V_{K+1}(x) - V_K(x) \leq \sum_{y \in \mathcal{X}} p(x, R_{K+1}(x), y) (V_K(y) - V_{K+1}(y))$$

Since $V_K(y) - V_{K+1}(y) \leq M_K$ for all $y \in \mathcal{X}$ and $\sum_{y \in \mathcal{X}} p(x, R_{K+1}(x), y) = 1$

We have $V_{K+1}(x) - V_K(x) \leq M_K \quad \forall x \in \mathcal{X}$

Which implies $M_{K+1} \leq M_K$.

Other one is similar (take $n=K+1$ in $*$ and $n=K$ in $**$)

Def A MDP is **STRONGLY APERIODIC** if under each policy π

$R: \mathcal{X} \rightarrow \mathcal{A}$ and for each state $x \in \mathcal{X}$, $p(x, R(x), x) > 0$

$\Rightarrow$ Any MDP that is not strongly aperiodic can be made into one by the following transformation:

ORIGINAL $\mathcal{X} \quad \mathcal{A} \quad r(x, a) \quad p(x, a, y)$ pick $\tau \in (0, 1)$
 TRANSFORMED $\bar{\mathcal{X}} = \mathcal{X} \quad \bar{\mathcal{A}} = \mathcal{A} \quad \bar{r} = r \quad \bar{p}(x, a, y) = \begin{cases} \tau p(x, a, y) & x \neq y \\ \tau p(x, a, y) + (1-\tau) & x = y \end{cases}$

What happens is: the transition probabilities conditioned on leaving state remain identical. Sojourn time in state x becomes geometric. Balance equations DON'T CHANGE:

ORIGINAL: $\pi(x) = \sum_{y \in \mathcal{X}} \pi(y) p(y, R(y), x)$

TRANSFORMED: $\pi(x) = \sum_{y \in \mathcal{X} \setminus \{x\}} \pi(y) \tau p(x, R(x), y) + \pi(x) (\tau p(x, R(x), x) + 1 - \tau)$

$\hookrightarrow$ divide by τ , get ORIGINAL.

Lemma

Consider a MDP that is unichain & strongly aperiodic with finite state space $|X| = N$. For any two types of policies

$(R_{N-1}, \dots, R_0)$ and $(Q_{N-1}, \dots, Q_0)$

and any two initial states x_1 and x_2 there exists a state y such that

$$P(X_N = y | X_0 = x_1, A_t = R_t(X_t)) > 0$$

$$P(X_N = y | X_0 = x_2, A_t = Q_t(X_t)) > 0.$$

Proof By contradiction on the unichain assumption (assume $S(k) \cap T(k) = \emptyset$)